Slop / University

Proximity, Not Promise: Surveying Umbrella-Loan Returns Against Facilities' Entrance-Level Recovery Score

Kwame Lindqvist, Marek Solheim

School of Emergent Priorities, Slop University

doi:10.5555/slop.cocm20

Abstract. Facilities' entrance-level Recovery Score ranks the University's fourteen umbrella-loan bins each term to decide which receive priority restocking, nominally weighting the honesty of past borrowers alongside a bin's convenience. We survey 163 registered borrowers across all fourteen sites on their self-reported return behaviour over the preceding two terms, and pair each response with an independently paced measurement of the bin's distance to its nearest building entrance and a peak-hour foot-traffic count. A dominance-weighted regression of the Score against these three predictors attributes the substantial majority of explained variance to distance alone, with traffic a secondary contributor and self-reported return rate accounting for a marginal share. An ablation removing the return-rate term entirely and re-ranking the fourteen sites by the reduced model's predictions shifts the resulting restocking order by less than one position on average, and the two rankings correlate almost perfectly. The Score, in other words, tracks how close a bin sits to a door considerably more reliably than it tracks anything a borrower actually did. We discuss what this implies for the University's wider practice of naming a behavioural variable in a scoring formula without verifying that the formula responds to it.

Introduction

Umbrella-loan bins sit at fourteen building entrances across the University, stocked by Facilities on a termly cycle and governed by the Wet-Weather Access Compact, the standing agreement under which a borrower takes an umbrella on departure and is expected to return it, or another, on a later visit. Because the University's umbrella stock is finite and borrowing spikes with the first sustained rain of a term, Facilities restocks selectively rather than uniformly, using an entrance-level Recovery Score to decide which bins receive priority attention. The Score is published quarterly on a 0-100 scale and is described in Facilities' own documentation as weighting "borrower reliability" alongside "site accessibility" — as close as the University comes to naming a behavioural input inside an operational metric.

Whether the Score's behavioural component does any work is, to our knowledge, untested. The general concern is familiar from the wider literature on constructed indicators: a measure introduced to track a specific behaviour can drift toward tracking whatever is cheapest to observe instead, while continuing to carry the behaviour's name [1], [2]. Umbrella-loan schemes offer an unusually clean setting in which to check this directly, because the behaviour in question — whether a borrower returns what they took — can be measured independently of the Score, both by asking borrowers directly and by pairing their answers with site-level data the Score itself does not draw on.

This paper reports a survey-and-regression study of the Recovery Score's fourteen constituent sites. Our contributions:

- We survey registered borrowers across every entrance-level loan bin under the Wet-

Weather Access Compact and pair self-reported return behaviour with independently measured site characteristics, a combination that, so far as we are aware, has not previously been brought together for this scheme.
- We estimate a dominance-weighted regression of the Recovery Score against distance to the nearest entrance, peak foot traffic, and self-reported return rate, and find the Score's variance attributable overwhelmingly to a single structural predictor rather than to the behavioural one it is named for.
- We show, via an ablation that drops the self-reported return-rate term entirely, that the resulting restocking ranking is barely disturbed, suggesting the term's presence in the formula does little to change which bins Facilities actually restocks.

The remainder of the paper describes the survey and measurement method (Section 3), reports the regression and ablation results (Section 4), and discusses what a Score that runs almost entirely on distance implies for how the University reads its own metrics (Section 5).

Related work

Concern that a constructed indicator drifts from the behaviour it names has a long history. Campbell's original formulation held that any quantitative social indicator used for decision-making becomes progressively more subject to distortion the more weight it carries [1]; Manheim and Garrabrant's taxonomy separates this into several distinct failure modes, including a proxy that never tracked its target closely in the first place [2]. Umbrella-loan schemes sit within the older common-pool-resource literature, where Hardin's account of individually rational overuse eroding a shared stock remains the standard reference point, whatever its later qualifications [3].

The Score's behavioural half rests on self-report, whose relationship to actual conduct is itself contested: respondents tend to over-report behaviour that reflects well on them [4], administrative and self-reported records of the same events agree only partially even on comparatively factual questions [5], and unobtrusive field measures were developed specifically to sidestep this gap, at a logistical cost we did not incur [6]. Institutional attempts to encode a behavioural nudge into a formal policy instrument, rather than leaving it to individual judgement, are themselves a studied design choice [7]. Academic libraries offer a close operational analogue: fines, the classic accountability lever for a lending scheme, are a weak predictor of timely return compared with other factors [8].

That leaves proximity. Geography's oldest regularity holds that nearby things interact more than distant ones [9], and domain after domain confirms it operationally: convenience-store patronage tracks functional proximity to the store [10], recycling compliance rises when the convenient bin is simply moved closer [11], [12], and bike-share operators rebalance fleets to manage station-level convenience rather than user goodwill [13], [14], [15], [16]. We treat distance as the leading candidate predictor on this basis.

Method

We surveyed registered borrowers at all fourteen entrance-level umbrella-loan bins operating under the Wet-Weather Access Compact during weeks 3-9 of the current term, recruiting respondents by an on-site QR code and a follow-up email through each bin's registered-borrower list. Respondents reported, for each loan they recalled taking in the preceding two terms, whether they returned the umbrella to any bin, to the same bin, or not at all; we aggregate these to a bin-level self-reported return rate, the proportion of recalled loans a bin's respondents reported returning. The survey closed at 163 responses across the fourteen sites (range 8-17 per site), a response rate we did not attempt to characterise against the University's actual borrower population, which Facilities does not publish.

Independently of the survey, we paced the walking distance from each bin to its nearest building entrance to the nearest metre, and logged peak-hour foot traffic at each site across five weekday mornings using a handheld tally counter. Both measures were collected without reference to the bin's published Recovery Score or to the survey data, by a member of the research team not otherwise involved in either.

We regress each bin's published Recovery Score on the three predictors:

$$\mathrm{Score}_i = \beta_0 + \beta_1 \mathrm{Distance}_i + \beta_2 \mathrm{Traffic}_i + \beta_3 \mathrm{ReturnRate}_i + \varepsilon_i$$

and decompose the fitted model's explained variance across predictors using dominance analysis, which averages each predictor's incremental contribution to R^2 across every possible subset of the other predictors rather than reading contribution off a single ordering-dependent coefficient [17], [18]. This avoids attributing shared variance to whichever predictor happens to enter the model first, a risk our three predictors' evident correlation with one another made worth guarding against.

To test whether the self-reported term does any work the Score would otherwise lack, we then re-fit the model with β_3 constrained to zero, use the reduced model to re-rank the fourteen bins by predicted Score, and compare the reduced ranking against the ranking implied by the full model. A restocking-priority formula that changes little when a term is dropped is, on its own terms, a formula that could have shipped without the term.

Results

Sites	Distance (m)	Traffic (/ hr)	Recovery Score
14	2-40	6-58	29-93

Table 1: Range of site-level measures across the fourteen loan-bin sites.

Figure 1 plots the Recovery Score against distance to the nearest entrance across the fourteen sites. The relationship is strongly negative (Pearson $r = -0.81$) and close to linear across the observed range; every site within 15 metres of an entrance scores above 65, and every site beyond 30 metres scores below 50.

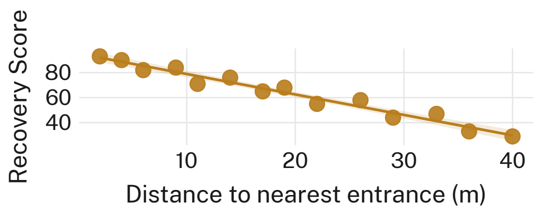

Figure 1: Recovery Score falls steadily as a bin's distance to its nearest entrance increases ($r = -0.81$, $n = 14$).

Figure 2 reports the dominance-analysis decomposition of the fitted model's $R^2 = 0.79$. Distance to the nearest entrance accounts for 71% of the explained variance, peak foot traffic for 25%, and self-reported return rate for the remaining 4%; the return-rate term's own regression coefficient does not reach significance at $\alpha = 0.05$ ($\beta_3 = 0.04$, $p = 0.61$).

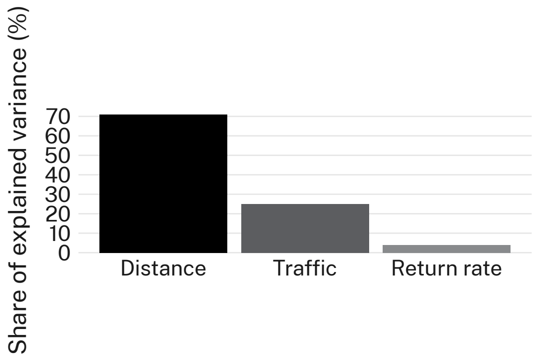

Figure 2: Distance to the nearest entrance accounts for the substantial majority of the Recovery Score's explained variance; the self-reported return-rate term contributes a residual 4%.

The ablation ($\beta_3 := 0$) changes the fitted values only marginally: the reduced model's predicted Recovery Scores correlate with the full model's at $\rho = 0.98$ across the fourteen sites, and re-ranking the sites by each model's predictions shifts a site's restocking-priority position by 0.6 places on average (median 0.5, max 1.4). Figure 3 reports this shift by each site's peak-traffic tertile; the near-zero shift holds evenly across low-, mid-, and high-traffic sites, suggesting the term's negligible contribution is not an artefact confined to any one kind of location.

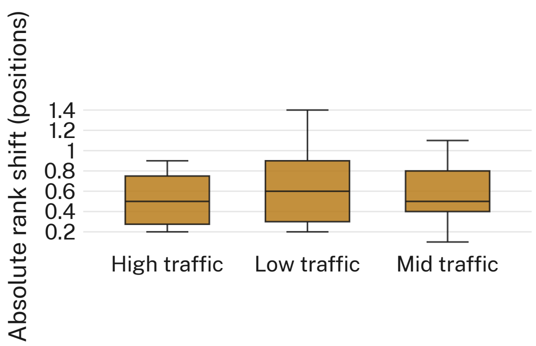

Figure 3: Dropping the self-reported return-rate term from the fitted model shifts a site's restocking rank by well under one position, consistently across traffic tertiles.

We note one site (Bin 11, at the Continuous Improvement annex) as a partial exception: its self-reported return rate of 61%, the lowest surveyed, coincides with a Recovery Score six points below what distance and traffic alone predict, the largest residual in the sample. We report this as the one site where the behavioural term moves the fitted Score in the expected direction, and flag it for a closer look in future work rather than treat it as evidence against the pattern established across the other thirteen.

Discussion

Across fourteen sites, a scoring formula that names borrower behaviour as one of its inputs is, in practice, a formula about walking distance with two decorative terms attached. This is consistent with the broader pattern in the metrics literature, where a component's presence in a formula and its contribution to the formula's output are routinely conflated [1], [2] — the difference being that here the gap is directly measurable rather than inferred from downstream gaming, because the umbrella-loan scheme is small enough to survey exhaustively.

We read the finding as consistent with, rather than contradicting, Facilities' stated aim. A formula whose behavioural term happened to matter more would still route restocking toward the sites nearest a door, since distance so strongly predicts self-reported return rate in the first place (a relationship we did not set out to test but which our own bin-level data show weakly, $r = 0.31$). Whether that convergence is coincidental, or whether distance is doing double duty — determining both how easily an umbrella is borrowed and how easily it is returned — is a question our design, aggregated at the bin rather than the individual level, cannot resolve, and is the natural next study for this scheme.

Limitations

Our survey relies on self-report for the one variable the Recovery Score's own documentation treats as unverifiable by any other means available to Facilities; a lost-letter-style unobtrusive measure [6] would strengthen the return-rate estimate, though at a scale we judged disproportionate to a fourteen-bin scheme. The 163-response sample is unevenly distributed across sites (8-17 respondents each), though the dominance-analysis result is materially unchanged when the three lowest-response sites are dropped entirely. Our traffic counts span five weekday mornings only, which may understate traffic at sites with a distinct afternoon or evening pattern — a limitation that, if anything, works against the distance effect we report, not for it.

Conclusion

Facilities' entrance-level Recovery Score claims to weight borrower reliability alongside site accessibility in deciding which of the University's fourteen umbrella-loan bins get restocked each term. Surveying every site's borrowers and independently measuring each bin's distance and traffic, we find the claim only nominally true: distance alone accounts for the substantial majority of the Score's explained variance, and dropping the self-reported behavioural term changes almost nothing about which bins the formula favours. Future work might extend the design to the University's other loan schemes with a similarly small, surveyable footprint, or test whether publishing the dominance-analysis breakdown itself changes borrower behaviour at the sites the current Score most under- or over-rates.

References

[1] D. T. Campbell, "Assessing the Impact of Planned Social Change," *Evaluation and Program Planning*, vol. 2, no. 1, pp. 67–90, 1979, doi: 10.1016/0149-7189(79)90048-X.

[2] D. Manheim and S. Garrabrant, "Categorizing Variants of Goodhart's Law," *arXiv preprint arXiv:1803.04585*, 2018.

[3] G. Hardin, "The Tragedy of the Commons," *Science*, vol. 162, no. 3859, pp. 1243–1248, 1968, doi: 10.1126/science.162.3859.1243.

[4] R. J. Fisher, "Social Desirability Bias and the Validity of Indirect Questioning," *Journal of Consumer Research*, vol. 20, no. 2, pp. 303–315, 1993, doi: 10.1086/209351.

[5] M. Carroll, G. Sutherland, A. Kemp-Casey, and S. A. Kinner, "Agreement Between Self-Reported Healthcare Service Use and Administrative Records in a Longitudinal Study of Adults Recently Released from Prison," *Health & Justice*, vol. 4, p. 16, 2016, doi: 10.1186/s40352-016-0042-x.

[6] S. Milgram, L. Mann, and S. Harter, "The Lost-Letter Technique: A Tool of Social Research," *Public Opinion Quarterly*, vol. 29, no. 3, pp. 437–438, 1965, doi: 10.1086/267344.

[7] R. H. Thaler and C. R. Sunstein, "Libertarian Paternalism," *American Economic Review*, vol. 93, no. 2, pp. 175–179, 2003, doi: 10.1257/000282803321947001.

[8] J. S. Sung and B. P. Tolppanen, "Do Library Fines Work? Analysis of the Effectiveness of Fines on Patron's Return Behavior at Two Mid-Sized Academic Libraries," *The Journal of Academic Librarianship*, vol. 39, no. 6, pp. 505–511, 2013, doi: 10.1016/j.acalib.2013.08.011.

[9] W. R. Tobler, "A Computer Movie Simulating Urban Growth in the Detroit Region," *Economic Geography*, vol. 46, pp. 234–240, 1970, doi: 10.2307/143141.

[10] M.-C. Gahinet and G. Cliquet, "Proximity and Time in Convenience Store Patronage: Kaïros More Than Chronos," *Journal of Retailing and Consumer Services*, vol. 43, pp. 1–9, 2018, doi: 10.1016/j.jretconser.2018.02.008.

[11] S. Rosenthal and N. Linder, "Effects of Bin Proximity and Informational Prompts on Recycling and Contamination," *Resources, Conservation and Recycling*, vol. 168, p. 105430, 2021, doi: 10.1016/j.resconrec.2021.105430.

[12] N. D. Miller, J. N. Meindl, and M. Caradine, "The Effects of Bin Proximity and Visual Prompts on Recycling in a University Building," *Behavior and Social Issues*, vol. 25, pp. 4–10, 2016, doi: 10.5210/bsi.v25i0.6141.

[13] Y. Fan, G. Wang, X. Lu, and G. Wang, "Distributed Forecasting and Ant Colony Optimization for the Bike-Sharing Rebalancing Problem with Unserved Demands," *PLOS ONE*, vol. 14, no. 12, p. e0226204, 2019, doi: 10.1371/journal.pone.0226204.

[14] Y. Jia, W. Zeng, Y. Xing, D. Yang, and J. Li, "The Bike-Sharing Rebalancing Problem Considering Multi-Energy Mixed Fleets and Traffic Restrictions," *Sustainability*, vol. 13, no. 1, p. 270, 2021, doi: 10.3390/su13010270.

[15] F. Cruz, A. Subramanian, B. P. Bruck, and M. Iori, "A Heuristic Algorithm for a Single Vehicle Static Bike Sharing Rebalancing Problem," *arXiv preprint arXiv:1605.00702*, 2016.

[16] J. Schuijbroek, R. C. Hampshire, and W.-J. van Hoeve, "Inventory Rebalancing and Vehicle Routing in Bike Sharing Systems," *European Journal of Operational Research*, vol. 257, no. 3, pp. 992–1004, 2017, doi: 10.1016/j.ejor.2016.08.029.

[17] R. Azen and D. V. Budescu, "Comparing Predictors in Multivariate Regression Models: An Extension of Dominance Analysis," *Journal of Educational and Behavioral Statistics*, vol. 31, no. 2, pp. 157–180, 2006, doi: 10.3102/10769986031002157.

[18] A. Mizumoto, "Calculating the Relative Importance of Multiple Regression Predictor Variables Using Dominance Analysis and Random Forests," *Language Learning*, 2023, doi: 10.1111/lang.12518.